

Lung Cancer: NSCLC & SCLC Management

Molecular Drivers, Immunotherapy Integration, and Surgical Criteria

Doc #08 of 15	Domain: Oncology & Cancer Treatment	Audience: Medical Professionals
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1. Epidemiology & Histology

Lung cancer remains the leading cause of cancer mortality globally (~1.8 million deaths/year). NSCLC accounts for ~85% of cases (adenocarcinoma 40%, squamous cell 25-30%, large cell 10-15%), SCLC ~15%. Risk factors: tobacco smoking (attributable fraction ~80%), radon exposure, occupational carcinogens (asbestos, arsenic), air pollution, and EGFR mutation (never-smokers, East Asian female).

2. Molecular Profiling — NSCLC Actionable Alterations

Alteration	Frequency (Adeno)	Preferred TKI	CDx Required
EGFR exon19del/L858R	~15% (EU/US); 40-50% (Asia)	Osimertinib (FLAURA2)	cobas EGFR / FoundationOne
ALK rearrangement	~5%	Alectinib or Lorlatinib	Vysis ALK FISH / D5F3 IHC
ROS1 rearrangement	~2%	Crizotinib or Entrectinib	FISH or NGS
KRAS G12C	~13%	Sotorasib or Adagrasib (2L)	FoundationOne CDx
MET exon14 skip	~3%	Tepotinib or Capmatinib	FoundationOne / RNA NGS

RET fusion	~1-2%	Selpercatinib or Pralsetinib	FoundationOne / FISH
BRAF V600E	~2-3%	Dabrafenib + Trametinib	THxID BRAF assay
NTRK1/2/3 fusion	<1%	Larotrectinib or Entrectinib	NGS / FISH
HER2 exon20 ins	~2-3%	T-DXd (trastuzumab deruxtecan)	NGS
EGFR exon20 ins	~2%	Amivantamab + Lazertinib	NGS

3. Stage I-III NSCLC — Curative Strategies

- Stage I: Lobectomy (gold standard) or SBRT for medically inoperable (BED ≥ 100 Gy10); segmentectomy non-inferior to lobectomy for ≤ 2 cm peripheral tumours (JCOG0802)
- Stage II: Resection + adjuvant osimertinib (if EGFR-mut; ADAURA) or atezolizumab (if PD-L1 $\geq 1\%$; IMpower010) or chemotherapy
- Stage IIIA resectable: Surgery + adjuvant chemo $\pm$ RT; consider neoadjuvant nivolumab + chemo (CheckMate 816) for pCR benefit
- Stage IIIB/C unresectable: Concurrent chemo-RT (cisplatin/etoposide or carbo/paclitaxel) $\rightarrow$ consolidation durvalumab x1 year (PACIFIC trial)

4. Stage IV Driver-Negative NSCLC

PD-L1 TPS	First-Line Regimen	Regimen Detail
$\geq 50\%$	Pembrolizumab monotherapy	KEYNOTE-024; OS benefit vs. chemo
1-49%	Pembrolizumab + platinum doublet	KEYNOTE-189 (non-sq) / KEYNOTE-407 (sq)
<1%	Platinum doublet + pembro or atezolizumab	Bevacizumab-atezolizumab-carbo-paclitaxel (IMpower150) for non-sq
Any, squamous	Carbo/paclitaxel + pembro or nivo+ipi	KEYNOTE-407 or CheckMate-9LA

5. SCLC Management

- Limited disease (LD-SCLC): Concurrent cisplatin/etoposide + thoracic RT (45 Gy BID or 60-66 Gy QD) $\rightarrow$ prophylactic cranial irradiation (PCI) if complete/good partial response
- Extensive disease (ED-SCLC): Carboplatin + etoposide + atezolizumab (IMpower133) or durvalumab (CASPIAN) $\rightarrow$ maintenance immunotherapy; lurbinectedin 2nd line
- Median OS LD-SCLC ~25 months; ED-SCLC ~12-13 months with immunotherapy
- Consider thoracic RT consolidation in ED-SCLC (CREST trial — PFS benefit); brain MRI surveillance in lieu of PCI for extensive disease

■ **MOLECULAR TESTING: All advanced NSCLC (non-squamous and squamous in never/light smokers) must undergo reflex broad NGS panel + PD-L1 IHC before initiating systemic therapy. Test concurrently, not sequentially, to avoid treatment delays.**